Cellular Regulators Underlying Skin Aging

Introduction

Aging can **weaken skin integrity**, reduce its ability to protect against external pathogens, and **impair wound healing.**

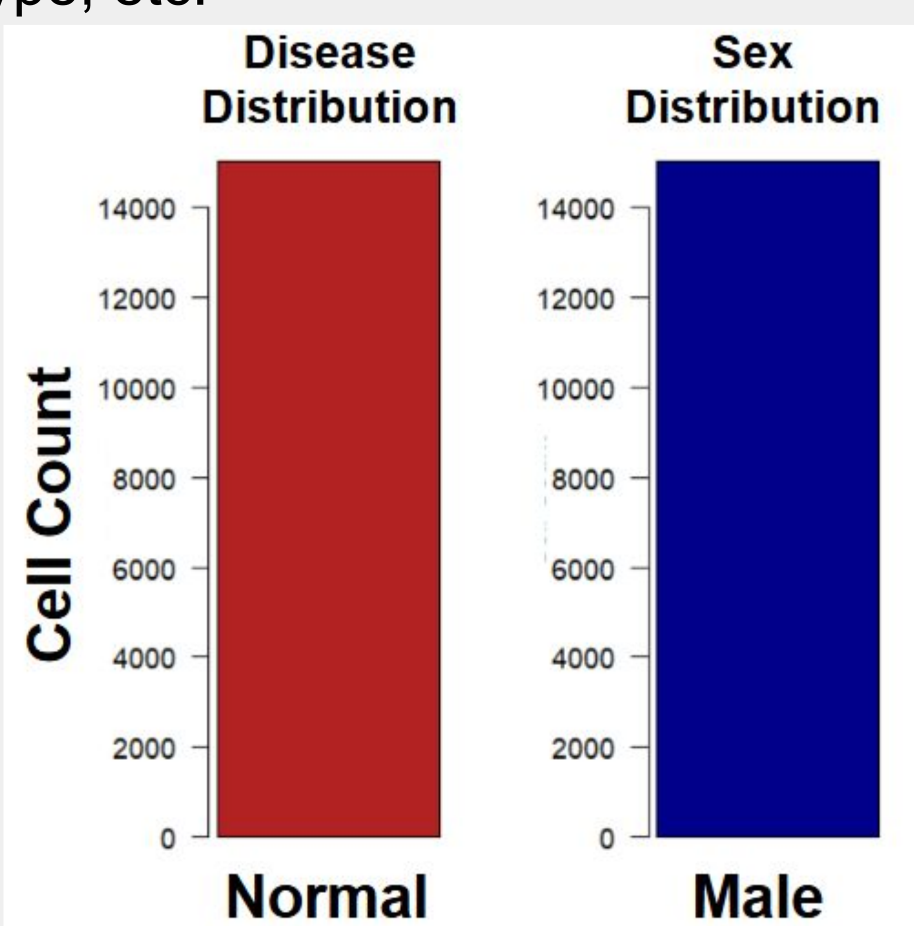
Using **single-cell RNA** sequencing datasets, we can identify which transcription factors (TF) **regulate** these changes.

Objectives

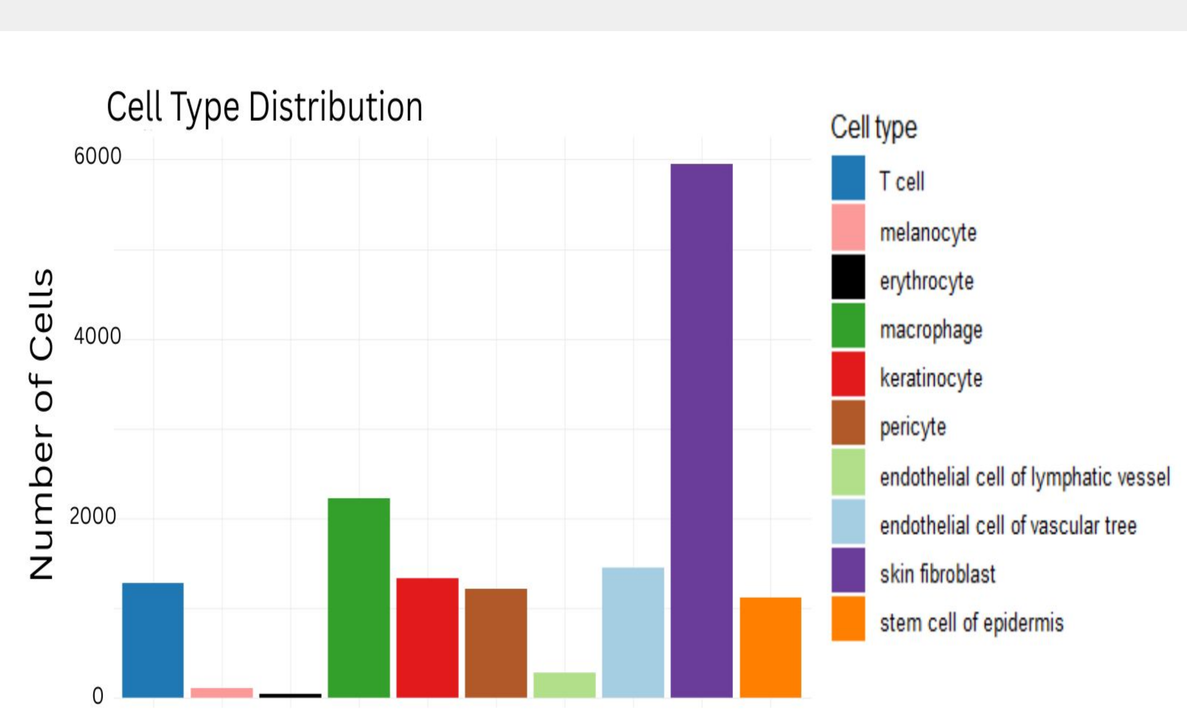
- Identify **top** TFs that control **upregulated** and **downregulated** genes (because of aging)
- Identify grouped targets for possible **therapeutics**

Dataset

- CellxGene (~15,000 cells)
- Input features: Health, age, genes, cell type, etc.

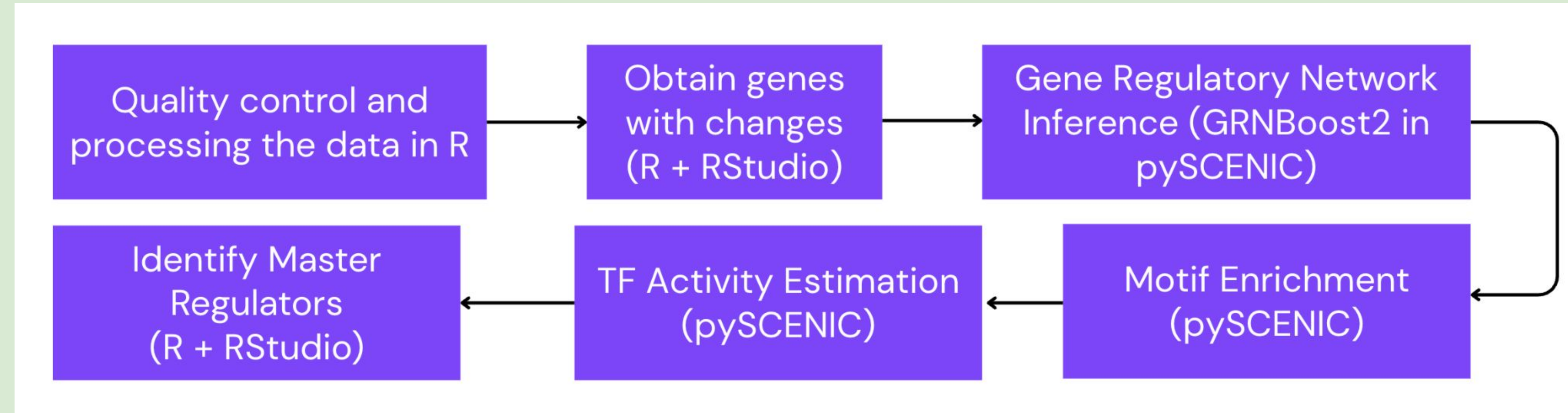


All cells being the same sex and disease classification **minimizes** confounding variables.

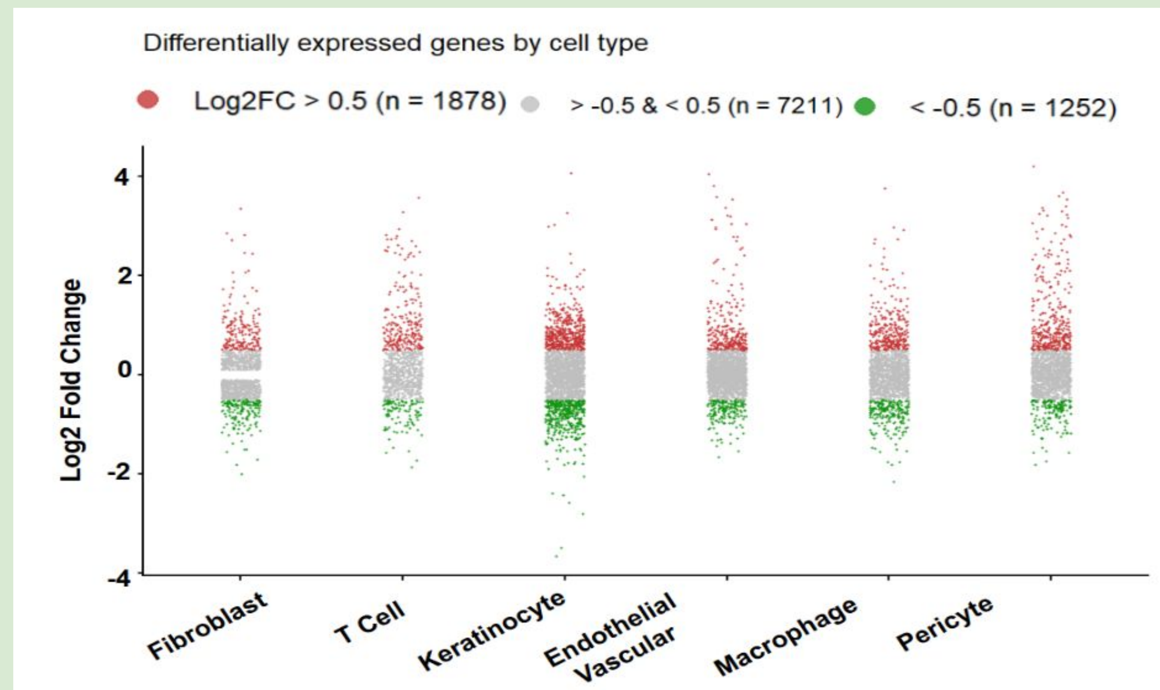


The makeup of the data through cell types.

Methodology



Obtain Gene Changes



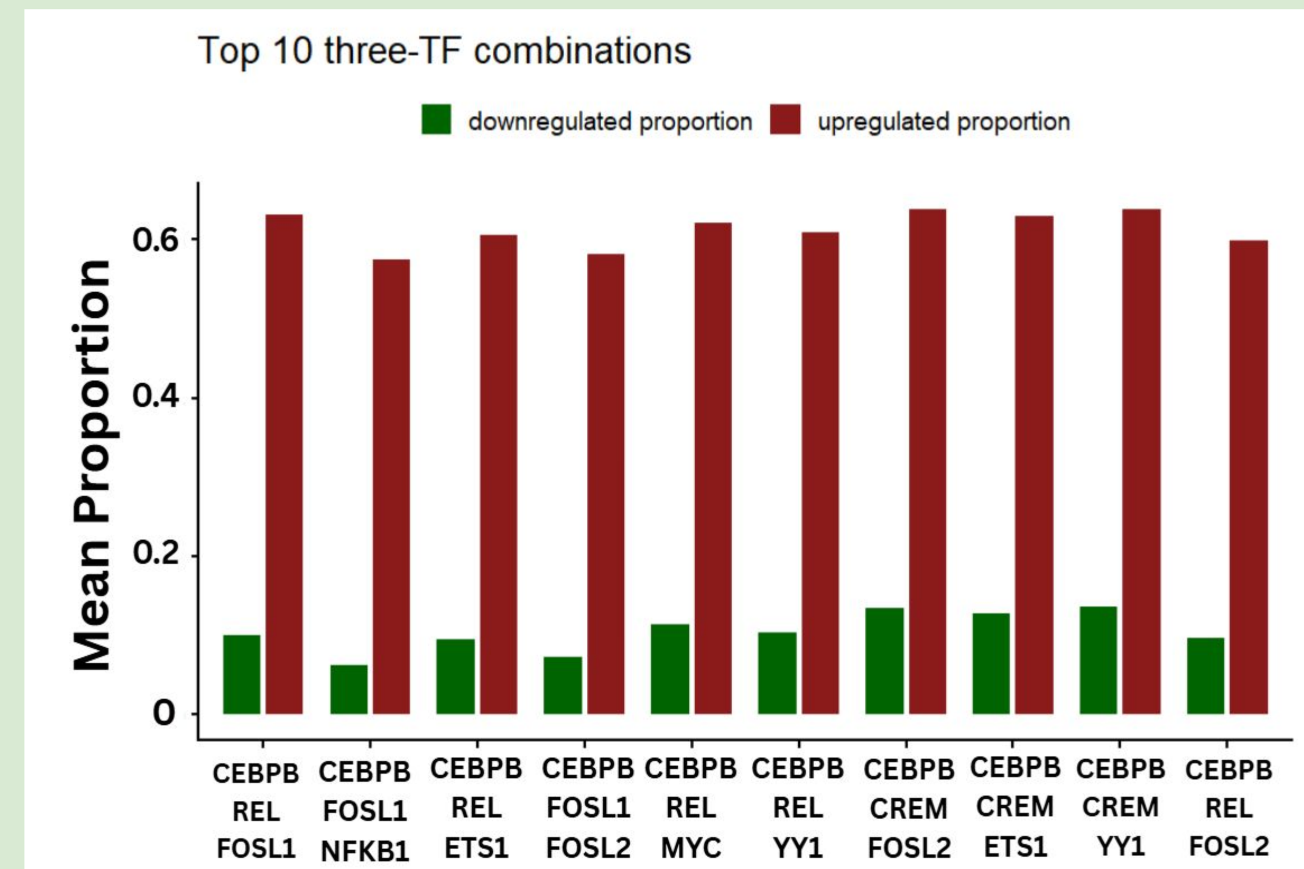
- Up regulated is logFC > 0.5 and p-val < 0.05
- Down regulated is logFC < -0.5 and p-val < 0.05

Master Regulators

CEBPB Score: 0.342	YY1 Score: 0.176
REL Score: 0.235	FOSL2 Score: 0.153
CREM Score: 0.226	CEBPD Score: 0.147
FOSl1 Score: 0.211	MYC Score: 0.141
NFKB1 Score: 0.178	ETS1 Score: 0.114

- Determine how many upregulated or downregulated genes are in each cell type
- Calculate the number of genes the TF targets
- Calculate a score:
  (up targeted/ up total) -
  (down targeted/ total down)

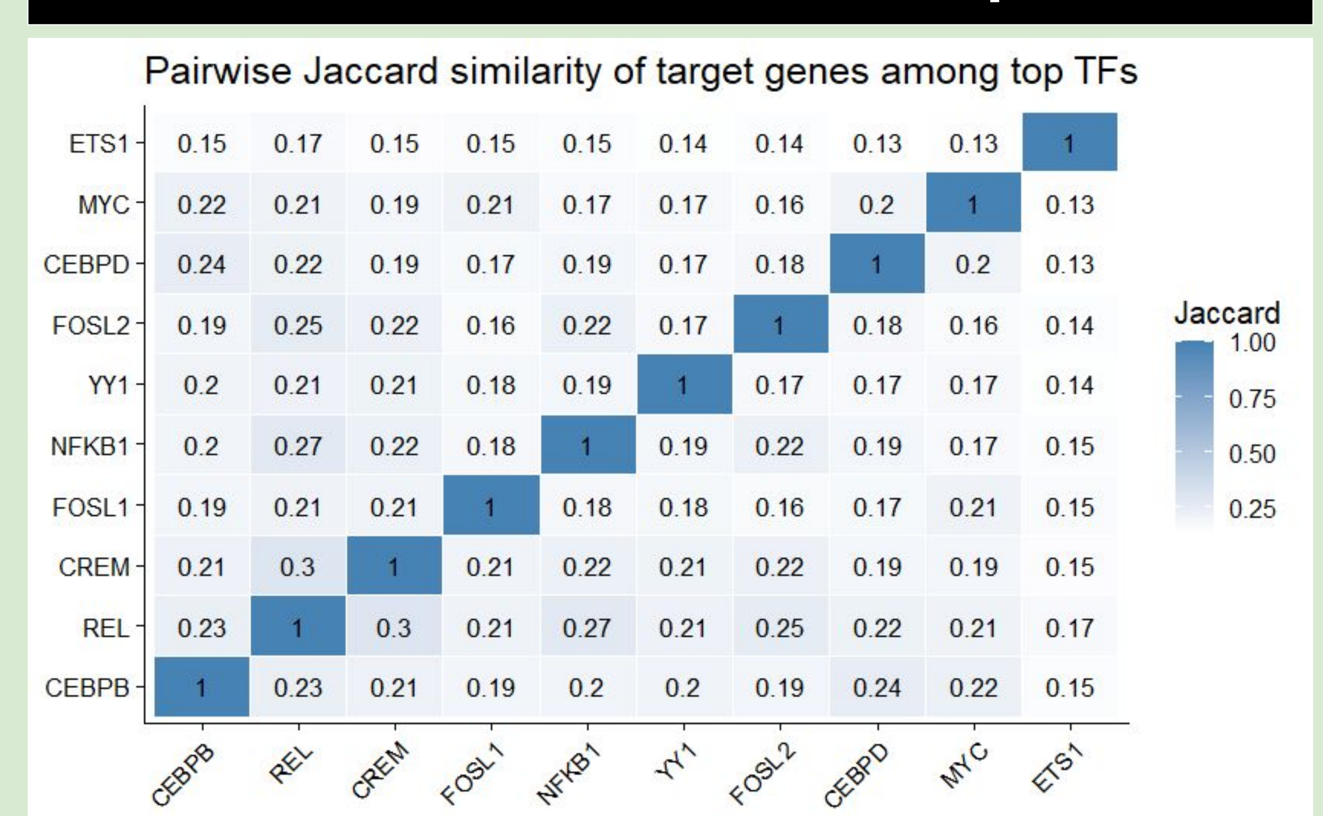
Grouping



Checked all groups of 3 TFs and calculated coverage (with overlap considered)

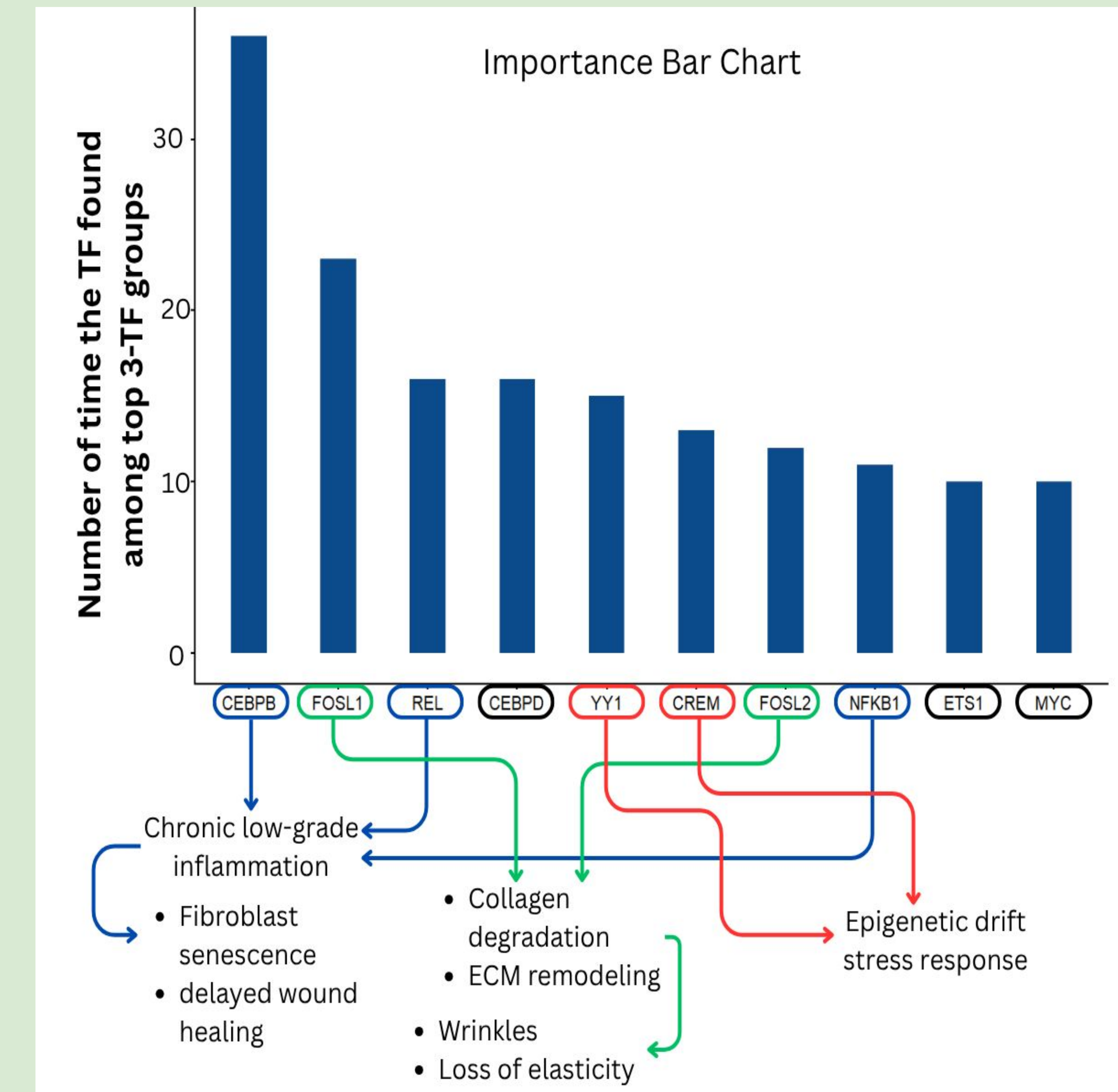
- Top 10 shows TFs to target for possible therapeutics

Jaccard Overlap



- Shows minimal overlap
  - Max overlap = 0.3 (disregarding matches)
  - Min overlap = 0.13
- In therapeutics, these targets are valid

Results



- Our analysis shows that grouping CEBPB, REL, and NFKB1 and targeting these TFs as a group can be a potential therapeutic approach for chronic low-grade inflammation and overcoming fibroblast senescence-mediated delayed wound healing.
- Additionally, FOSL1 and FOSL2 can be targeted to overcome collagen degradation in ECM remodeling, which is typically associated with wrinkles.
- Furthermore, a treatment approach targeting YY1 and CREM as a group can potentially overcome epigenetic drift stress response.

Conclusion

The identified TFs regulate around 60% of genes found to be upregulated in the skin during the aging process, causing chronic inflammation and altering extracellular matrix organization, which can represent novel therapeutic targets. Our analytical approach can be tailored to identify novel disease-related genetic variations and suggest potential therapeutic options.

References

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2- Shin, SH, et al. "Skin Aging from Mechanisms to Interventions: Focusing on Dermal Aging." Frontiers in Physiology, vol. 14, 2023, 1195272. doi:10.3389/fphys.2023.1195272.

3- Jin, Shizuo, et al. "Inference and Analysis of Cell-Cell Communication Using CellChat." Nature Communications, vol. 12, no. 1, 2021, 1088. doi:10.1038/s41467-021-21246-9.